Appendix Group 2

Sources

file	row id	column	value	source type	source or calculation	sanity check
products.csv	SPK_01	mass_kg	0.93	public	JBL manufacturer specification; accessed 2026-11-05. https://da.jbl.com/on/demandware.static/-/Sites-masterCatalog_Harman/default/dw0a39d513/pdfs/JBL_Flip_6_SpecSheet_English.pdf	930 g net; model BOM excluding packaging sums to 0.66 kg.
products.csv	SPK_01	list_price_eur	117.64	public	JBL DE displayed gross price EUR 139.99; net comparison = 139.99/1.19 = 117.64. Page marked sold out. Accessed 2026-11-05. https://da.jbl.com/JBLFLIP6BLKEU.html	Gross reference price is not manufacturer revenue; 19% VAT convention; Margin check is illustrative.
bom.csv	SPK-SYEMP-HOUSING	qty_per_unit	1	guesstimate	One complete functional kit per speaker, subparts grouped.	300 speakers/week requ

Key Arguments of Papers:

1. Temporary sourcing diversification (Whitney et al., 2014)

- The authors use historical case studies to investigate how product and production specificity affect the feasibility and success of temporary alternative sourcing during supply chain disruptions.

- Their analysis reveals that alternative sourcing options only support recovery if the components do not require excessively specialized manufacturing capabilities.

- For the project, this implies that a shortened lead time through a backup supplier is only effective if they can actually supply compatible and qualified components in time.

2. Measuring sourcing resilience (Ivanov, 2025)

- Ivanov combines network analysis with discrete-event simulation in his work to evaluate how well supply chains can respond to unforeseen disruptions.

- His findings demonstrate that purely structural network data is insufficient, and only dynamic simulations can capture the actual temporal and operational effects of delays.

- Applied to the project, this justifies why the baseline, the disruption, and the countermeasures must be compared over time using metrics such as weekly output, costs, CO_2, and downtime, rather than simply assuming an alternative supplier network is flawless

AI Transparency Note

We used AI tools to support us during different stages of this project. First, AI helped us understand the project requirements more easily and helped us divide tasks among team members. We also used AI to search for relevant scientific papers and to help write and debug our Python code in Google Colab. To keep our work on track, we continuously used AI for troubleshooting and final checks after completing each project step.

Using AI also taught us an important lesson about double-checking results. Early on, an AI tool made a mistake while reading our dataset. Unaware of this, we built our very first simulation on these incorrect figures. Once we noticed the error, we corrected the data and re-ran the entire simulation from scratch. This showed us why human oversight is always necessary when working with AI.